

Integrated Library Management System (CDM Library)

Capstone Project Manuscript Reference & Technical Specification Document

TARGET CLIENT: COLEGIO DE MONTALBAN (CDM)

1. Executive Summary & Project Overview

The **Integrated Library Management System (CDM Library / cdmLib)** is a comprehensive, multi-platform academic library ecosystem engineered for Colegio de Montalban (CDM). The solution bridges campus library administration with student mobile access by unifying a desktop management panel, a native Android mobile application, a RESTful API service, and a public promotional web portal.

2. User Access & Control Matrix

User Role	Access Channel	Permissions & Core Responsibilities
Head Librarian / Admin	Desktop Web Panel	Full system control: staff account creation and approval, catalog CRUD operations, circulation override, fine management, and analytics reporting.
Staff Librarian	Desktop Web Panel	Daily desk operations: processing book checkouts/returns, managing active reservations, student profile inspection, and inventory stock monitoring.
Student / Faculty	Android Mobile App	24/7 catalog search, personal active loan tracking, return due-date alerts, book reservation submissions, and penalty balance viewing.
Guest / Public User	Web Portal	Zero-login live book search, library announcements, installation guides, and direct mobile app (.APK) downloads.

3. Core System Modules & Functional Features

3.1 Authentication & Security Subsystem

- Role-Based Access Control (RBAC):** Strict segregation of administrative, operational, and student privileges.

- **Staff Registration & Approval Workflow:** New librarian sign-ups remain pending until explicitly reviewed and approved by the Head Librarian.
- **Token-Based Authentication:** Secure session handling powered by JSON Web Tokens (JWT) with 24-hour validity.
- **Cryptographic Password Security:** Passwords are protected using PBKDF2 with SHA-512 key derivation and 16-byte random salts, evaluated via constant-time buffer comparisons to eliminate timing attack vectors.

3.2 Catalog & Book Inventory Management

- **Book Record CRUD:** Full management of book metadata including Title, Author, ISBN (unique constraint), Category, Abstract/Summary, Publication Year, Cover URL, and Quantity.
- **Dynamic Stock Allocation:** Automatic decrement of available inventory when books are checked out or reserved, and increment upon return.
- **Multi-Criteria Live Search:** Sub-second catalog queries filtered by title, author, category, or ISBN.
- **Circulation Frequency Tracker:** Automated metric tracking (`borrow_count`) to pinpoint high-demand academic resources for collection development.

3.3 Circulation & Loan Transaction Subsystem

- **Check-Out Module:** Links physical book copies to verified student profiles with assigned borrowing terms and due dates.
- **Check-In & Inspection Module:** Logs return timestamps, assesses book condition (*Good*, *Damaged*, or *Lost*), and updates inventory counts instantly.
- **Automated Penalty Calculation:** Dynamic fine assessment based on elapsed days past the return due date.
- **Transaction State Lifecycle:** Real-time status management tracking loans across states: *active*, *returned*, *overdue*, and *lost*.

3.4 Remote Book Reservation Module

- **Mobile Holding Requests:** Allows students to submit reservation requests for physical titles via their mobile devices.
- **Pickup Window Scheduling:** Assigns designated pickup target dates for accepted reservations.
- **Reservation Workflow Management:** Operational state transitions: *pending* → *fulfilled* / *cancelled*.

3.5 Borrower Records & Student Profile Subsystem

- **Centralized Student Directory:** Maintains student numbers, institutional email addresses, contact numbers, course/program (e.g., BSIT, BSCE, BSEd), and year levels.

- **Borrowing History Audit:** Complete transaction history detailing current loans, historical returns, active reservations, and accrued penalties.

3.6 Analytics & Executive Dashboard

- **Real-Time KPI Cards:** Instant visual indicators for Total Books, Total Available Copies, Active Loans, Overdue Books, Pending Reservations, and Total Revenue Fines.
- **Data Visualization Charts:** Embedded Recharts visualizations presenting loan volume trends, category distribution, and library activity.

4. Technical Architecture & Database Specifications

4.1 Software Stack Specification

Layer	Technologies Used
Frontend	React 19, TypeScript, Vite 6, TailwindCSS 4, Radix UI, Motion (Framer), Recharts, Lucide Icons
Backend API	Node.js (ES Modules), Express.js (v5), CORS middleware
Database	SQLite 3 via better-sqlite3 driver (PRAGMA foreign_keys = ON)
Mobile App	Native Android Application (.APK distribution)
Security	JSON Web Tokens (JWT), Node.js crypto (PBKDF2 SHA-512)

4.2 Database Schema Architecture

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1. librarians (id, first_name, last_name, email, phone, employee_id, role, username, password, status)
2. books (id, title, author, isbn, category, cover, abstract, available, total, borrow_count, publish_year)
3. transactions (id, book_id, book_title, student_name, student_id, librarian_name, borrow_date, due_date, return_date, status, book_condition, penalty)
4. reservations (id, book_id, book_title, student_name, student_id, reservation_date, pickup_date, status)
5. students (id, name, email, phone, course, year_level, status)
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5. Problems Solved & Operational Impact

Key Contributions to Institutional Efficiency:

1. **Removes Physical & Spatial Limits:** Students gain 24/7 access to catalog availability and remote reservations without physically queuing at the campus library.
2. **Mitigates Late Returns:** Automated due-date notifications reduce overdue instances and minimize student fine burdens.
3. **Streamlines Administrative Workflow:** Eliminates manual paper logging and reduces check-in/checkout processing time for librarians.
4. **Ensures Inventory Integrity:** Synchronized database operations ensure stock counts reflect physical availability at all times.